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Homework 8

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1. NEWMAN EXERCISE 6.4

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We look to model the circuit of resistors in Figure 1. The voltage source, V_+ provides 5 V. We can utilize Ohm's Law and Kirchoff current law to find the voltages at each junction. The result is a system of four equations. For the junction at V_1,

$$4V_1 - V_2 - V_3 - V_4 = V_+ \tag{1}$$

for the junction at V_2,

$$V_1 - 3V_2 + V_4 = 0 \tag{2}$$

for the junction at V_3,

$$V_1 - 3V_3 + V_4 = -V_+ \tag{3}$$

and for the junction at V_4,

$$V_1 + V_2 + V_3 - 4V_4 = 0. \tag{4}$$

We can combine these four equations into a matrix equation,

$$\begin{bmatrix} 4 & -1 & -1 & -1 \\ 1 & -3 & 0 & 1 \\ 1 & 0 & -3 & 1 \\ 1 & 1 & 1 & -4 \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \\ V_3 \\ V_4 \end{bmatrix} = \begin{bmatrix} V_+ \\ 0 \\ -V_+ \\ 0 \end{bmatrix} \tag{5}$$

where we can now employ linear algebra techniques to solve. We can use row reduction techniques to perform LU decomposition. This forms an upper triangular matrix that allows us to use backward substitution to solve for the unknown voltage values. We will create a python program with the numpy linear algebra solve method that employs this technique.

We begin with pseudo code:

```
import dependencies

initailize coefficient matrix
initialize ordinate values

np.linalg.solve

return values of V
```

We find that $V_1 = 3$ V, $V_2 = 1.67$ V, $V_3 = 3.33$ V, and $V_4 = 2$ V.

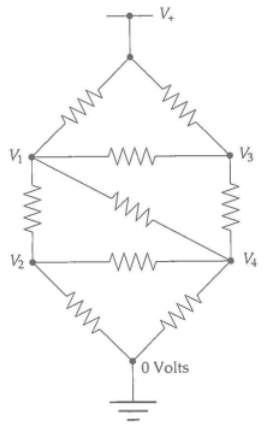

FIG. 1: Circuit of resistors.

2. NEWMAN EXERCISE 6.9

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We look to model an asymmetric quantum well. Suppose we have a particle of mass M in a one-dimensional quantum well of width L that isn't square. Let's also assume that the potential varies within the well. We note that the spatial part $\psi(x)$ of the wavefunction is an eigenfunction that obeys the time-independent Schrodinger equation $\hat{H}\psi(x) = E\psi(x)$, where the Hamiltonian operator $\hat{H}$ is given by

$$\hat{H} = \frac{-\hbar^2}{2M}\frac{\mathrm{d}^2}{\mathrm{d}x^2} + V(x). \tag{6}$$

Assuming the walls are infinitely high and the wavefunction is zero outside the well, the wavefunction can be expressed as a Fourier sine series,

$$\psi(x) = \sum_{n=1}^{\infty} \psi_n \sin\frac{\pi n x}{L} \tag{7}$$

where ψ_1, psi_2, ... are the Fourier coefficients. Starting with

$$\int_0^L \sin\frac{\pi m x}{L}\sin\frac{\pi n x}{L}\,dx = \begin{cases} \frac{L}{2} & \text{if } m = n \\ 0 & \text{otherwise} \end{cases} \tag{8}$$

we can rewrite this expression such that

$$\int_0^L \psi_m(x)\psi_n(x)\,dx = \frac{L}{2}\delta_{mn} \tag{9}$$

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where δ_{mn} is the Kronecker delta and is defined by

$$\delta_{mn} = \begin{cases} 1 & \text{if } m = n \\ 0 & \text{if } m \neq n. \end{cases} \tag{10}$$

Now, we want to consider

$$\sum_{n=1}^{\infty} \psi_n \int_0^L \sin \frac{\pi m x}{L} \hat{H} \sin \frac{\pi n x}{L} \, dx. \tag{11}$$

We can use the time-independent Schrodinger equation to substitute $E\psi_n$ for $\hat{H}\psi_n$ and the definitions of ψ_n and ψ_m to yield

$$\sum_{n=1}^{\infty} \psi_n E \int_0^L \psi_m(x)\psi_n(x) \, dx. \tag{12}$$

We know the value of this integral from Equation 9,

$$\sum_{n=1}^{\infty} \psi_n E \frac{L}{2} \delta_{mn}. \tag{13}$$

Finally, the definition of the Kronecker delta implies that this solution is non-trivial only when $n = m$, so as we sum across all ψ_n, only when we have ψ_m will the solution be non-zero. This shows that

$$\sum_{n=1}^{\infty} \psi_n \int_0^L \sin \frac{\pi m x}{L} \hat{H} \sin \frac{\pi n x}{L} \, dx = \frac{1}{2} L E \psi_m \tag{14}$$

We can construct a Hamiltonian matrix $\mathbf{H}$ with entries

$$\begin{aligned} H_{mn} &= \frac{2}{L} \int_0^L \sin \frac{\pi m x}{L} \hat{H} \sin \frac{\pi n x}{L} \, dx \\ &= \frac{2}{L} \int_0^L \sin \frac{\pi m x}{L} \left(\frac{-\hbar^2}{2M} \frac{\mathrm{d}^2}{\mathrm{d}x^2} + V(x) \right) \sin \frac{\pi n x}{L} \, dx. \end{aligned} \tag{15}$$

We will want to consider the case when $V(x) = ax/L$. Plugging this into Equation 15,

$$\begin{aligned} H_{mn} = &\frac{n^2 \pi^2 \hbar^2}{M L^3} \int_0^L \sin \frac{\pi m x}{L} \sin \frac{\pi n x}{L} \, dx \\ &+ \frac{2a}{L} \int_0^L x \sin \frac{\pi m x}{L} \sin \frac{\pi n x}{L} \, dx \end{aligned} \tag{16}$$

Note that the second integral

$$\int_0^L x \sin \frac{\pi m x}{L} \sin \frac{\pi n x}{L} \, dx \tag{17}$$

evaluates to

$$= \begin{cases} L^2/4 & \text{if } m = n. \\ -(\frac{2L}{\pi})^2 \frac{mn}{(m^2-n^2)^2} & \text{if } m \neq n \text{ and one is even, one is odd} \\ 0 & \text{if } m \neq n \text{ and are both even or odd} \end{cases} \tag{18}$$

Equation 16 then simplifies to

Great derivation of this

$$\begin{cases} \frac{n^2 \pi^2 \hbar^2}{2ML^2} + \frac{aL}{2} & \text{if } m = n \\ -(\frac{8aL}{\pi^2}) \frac{mn}{(m^2-n^2)^2} & \text{if } m \neq n \text{ and one is even, one is odd} \\ 0 & \text{if } m \neq n \text{ and are both odd or even} \end{cases} \tag{19}$$

We can create a python program that creates the matrix $\mathbf{H}$ with these defined elements. We will consider $L = 5$Å, $a = 10$eV, $q_e = 1.6022 \cdot 10^{-19}$J, and $M_e = 9.1094 \cdot 10^{-31}$kg. We will look to create an NxN matrix for $\mathbf{H}$ and compute the corresponding energy eigenvalues. We will also look to plot the probability density functions for the first three energy eigenstates for range of $0 \leq x \leq L$.

We begin with pseudo code

```
import dependencies

initialize constants

def H(m,n)
    if m = n:
        first case
    elif m != n and one odd and one even:
        second case
    else:
        third case

for i in range(N):
    for j in range(N):
        fill out H matrix

use numpy to solve for eigenvalues of H

use numpy to solve for eigenvectors of H

def psi(n, x):
    calculate wavefunction

plot probability densities
```

For a 10x10 matrix, we find that the ten energy eigenvalues are 5.83639118, 11.18113355, 18.66298345, 29.14436203, 42.65533201, 59.18562839, 78.72986474, 101.28614293, 126.85221998, and 155.55635795 eV.

For a 100x100 matrix, we find that the first ten energy eigenvalues are 5.83639077, 11.1811322, 18.6629816, 29.1443532, 42.6553229, 59.1855758, 78.7298129, 101.285512, 126.851388, and 155.426736 eV.

This seems to suggest that regardless of $\mathbf{H}$'s size, the accuracy in the value of each eigenvalue changes only marginally.

Figure 2 shows the probability densities for the ground, first excited, and second excited states.

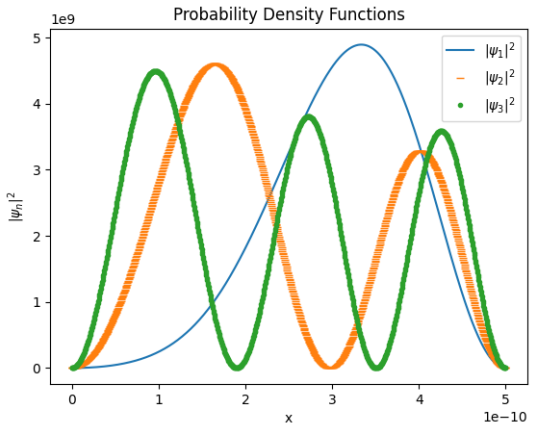

FIG. 2: Probability density functions for the ground, first excited, and second excited states.

From the plots, we can see how a non-zero potential in an asymmetric well causes our expectation of where the electron is to change compared to a square well. In a square well at the ground state, we'd expect it to be in the center. In this asymmetric well, we expect to the electron to be located more towards the right. Interestingly, as the electron's energy state increases, our expectation moves to the left significantly.

3. SURVEY

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This problem set took roughly 5-6 hours. Exercise 6.4 was a good chance to practice matrix operations within python. Exercise 6.9 was somewhat tedious with the latex typesetting and having to show the mathematics, however, was satisfying to complete.

Problem 6.4

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Computational Physics/Astrophysics, Winter 2023:

Grading Rubrics [1]

Haverford College, Prof. Daniel Grin

For coding assignments, roughly 25 points will be available per problem.

+3 1. Does the program complete without crashing in a reasonable time frame? If yes, up to +3 points.

+1 2. Does the program use the exact program files given (if given), and produce an answer in the specified format? If yes, +1 points

+2 3. Does the code follow the problem specifications (i.e numerical method; output requested etc.) Up to +2 points

+4 4. Is the answer correct? Up to+4points

+2 5. Is the code readable? Up to+2points

. 5.1. Are variables named reasonably?

. 5.2. Are the user-functions and imports used?

. 5.3. Are units explained (if necessary)?

. 5.4. Are algorithms found on the internet/book/etc. properly attributed?

[1] Inspired by rubric of D. Narayanan, U. Florida, and C. Cooksey, U. Hawaii

+3 6. Is the code well documented? +3points

. 6.1. Is the code author named?

. 6.2. Are the functions described and ambiguous variables defined?

. 6.3. Is the code functionality (i.e. can I run it easily enough?) documented?

+10 7. LaTeX writeup (up to 10 points)

- Are key figures and numbers from the problem given? (3 points)
- Is a brief explanation of physical context given? (2 points)
- If relevant, are helpful analytic scalings or known solutions given? (1 point)
- Are 3-4 key equations listed (preferably the ones solved in the programming assignment) and algorithms named? (2 points)
- Are collaborators clearly acknowledged? (1 point)
- Are any outside references appropriately cited? (1 point)

Note, even if (1), (2), (3), or (4) are not correct, one can still obtain many points via (5), (6), and (7).

Problem 6.9

Flanders +23

Computational Physics/Astrophysics, Winter 2023:

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Haverford College, Prof. Daniel Grin

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+3 1. Does the program complete without crashing in a reasonable time frame? If yes, up to +3 points.

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+2 3. Does the code follow the problem specifications (i.e numerical method; output requested etc.) Up to +2 points

+3 4. Is the answer correct? Up to+4points

Everything looks to be mirrored on the x-axis. The large blue peak should be on the left for example. Your energies and overall shape look good though.

+2 5. Is the code readable? Up to+2points

. 5.1. Are variables named reasonably?

. 5.2. Are the user-functions and imports used?

. 5.3. Are units explained (if necessary)?

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. 6.3. Is the code functionality (i.e. can I run it easily enough?) documented?

There is not much documentation in your code

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